

Lecture 2 In-Class Exercise

PART 1

Use the following command to read aggregate PISA scores for the 38 OECD countries for various years between 2000 and 2022 into Stata:

```
use https://github.com/spcorcor18/LP0-8800/raw/main/data/PISA-2000-2022.dta, clear
```

1. There is an indicator variable called *aggobs* that is equal to 1 for observations that represent aggregates (e.g., OECD average). We won't be using these, so you can drop these observations going forward (or condition later commands using *aggobs* $\sim=$ 1).
2. To see which countries are included in the dataset, provide a frequency distribution for the *jurisdiction* variable.
3. For each of the major test subjects (math, reading, science), provide a frequency distribution showing the count of observations with nonmissing country average test scores in each year.
4. For each major test subject **in 2022**, obtain the mean, median, and standard deviation for the country average test score. Interpret the standard deviation(s) in words. How do the mean and median compare for each subject?
5. For each major test subject **in 2022**, create a histogram for the country average test score. Describe the shape of these distributions.
6. For each major test subject **in 2022** obtain the skewness statistic. Does the statistic align with what you observed in the histograms? Would you say these distributions are "significantly skewed"?
7. There is a variable called *r_sd* containing the standard deviation of reading scores within a country. (There are similar variables for math and science). Which three countries had the highest within-country standard deviation in reading scores?
8. For each major test subject **in 2022**, provide a "five number summary" (min, Q1, median, Q3, max) for the country average test score and include the interquartile range (IQR). Create a boxplot for each major test subject. Can you interpret the quartiles in words? The IQR?
9. Create a combined box plot that shows the distribution of country average **math** scores by year. How would you describe what you see?

PART 2

Use the following link to download the file *PISA-2015-seven-countries*. (The file is too large for Github). This file contains *individual student* data from the PISA 2015 for seven countries, including the United States.

<https://vanderbilt.box.com/s/etiyvhr6t99pwum1f1l2v81j5ya1i5vf>

1. For this section, keep only students in the United States (see the *cnt* variable).
2. Find the survey item in which students are asked how many cell phones they have in their household. (Hint: use the `lookfor` command and a search term). Produce a frequency distribution for this variable using both `tabulate` and `fre`. How many observations are missing data for this item? Create a bar graph showing the distribution of responses for this item.
3. There is a continuous variable in the dataset called *escs* which is an index of economic, social, and cultural status. Generate a histogram for this variable.
4. Use the `xtile` command to place students into quartiles based on their *escs* index. Note this command will create a new variable. Verify that this command worked by looking at the min and max values of *escs* for students in each quartile.
5. If you have extra time, peruse the variables available in this dataset.